

MODULE 3.2

Modeling Bungee Jumping

Downloads

The text's website has the following files containing the models in this module available for download with various system dynamics tools: *VerticalSpring* and *Bungee*.

Introduction

On April Fool's Day in 1979, four members of Oxford University's Dangerous Sports Club, dressed in tails and top hats, climbed out onto the Clifton Suspension Bridge in Bristol, U.K. Each attached one end of a nylon-braided, rubber shock cord to himself and the other to the bridge. Then, they jumped off toward the 250-ft Avon Gorge. Voila! The sport of bungee jumping had begun in the western world.

What in the world possessed these men to do such a thing? The story goes that they watched a film on "land divers" from Pentecost Island in the South Pacific and became inspired to try diving themselves.

What are land divers? These divers are the male inhabitants of Pentecost who dive from platforms at various heights along a wooden tower. For these dives, lianas (vines) attached to the tower are tied to their ankles. Divers may be as young as 7 yr of age. Naturally, the lianas have to be selected very carefully. They must be just the right length and elasticity for the height of the platform and the weight of the diver. Consideration must be given to the length of the platform (which collapses and absorbs some shock), the slope of the land, and the swaying of the tower. A perfect dive will have the hair of the diver just brushing the ground. A miscalculation might be fatal. Land diving is part of ceremonies that ensure the yam harvest and fertility. Now, extreme-sports enthusiasts come from all over the world to experience land diving.

How did this practice get started? Why would men choose to jump from platforms with vines tied to their ankles? The annual land dives are based on local lore about a young girl betrothed to a much older man. The frightened young girl, at-

tempting to escape her new husband, climbed high into a banyan tree. The angry husband pursued her up the tree. As he ascended, she tied vines to her ankles and jumped. The husband followed, but without the vines, and was killed. Today's young men may prove that they have learned the escape trick and will not be fooled again (Menz 1993).

Physics Background

The action of a bungee cord is similar to that of a spring. Thus, in modeling bungee jumping we employ a critical law of physics concerning springs, Hooke's law.

Hooke's law pertains to springs that are perfectly elastic, so that they can "spring back" fully. Thus, the law can be applied only as long as the spring has not been stretched too much. The law states that within the elastic limit of the spring, a restoring force (F) applied to a spring is in proportion to and in the opposite direction of the spring's displacement (s) from its equilibrium position. Thus, for a spring constant k , which varies depending on the spring, we have the following formula for the restoring force:

$$F = -ks$$

Figure 3.2.1 illustrates the situations for stretched and compressed springs. When a resting spring is pulled or pushed, a force is exerted to restore the spring. As long as we do not stretch it beyond its elastic limit, the further we pull or push the spring, the more force going in the opposite direction results. For example, as we tug farther on an exercise spring, we feel more resistance.

Hooke's Law Within the elastic limit of a spring, where F is the applied force, k is the spring constant, and s is the displacement (distance) from the spring's equilibrium position, the following formula holds:

$$F = -ks$$

Quick Review Question 1

- If displacement is in meters, give the units of the spring constant.
- For a displacement of 0.1 m and a spring constant of 5 kg/s², give the restoring force along with its units.

Quick Review Question 2

This question reflects on Step 2 of the modeling process—formulating a model—for a vertical spring's length. Suppose a spring has a weight on the end that we pull

down or push up. Consider down as the positive direction. We simplify this first attempt at a model by ignoring friction and the weight of the spring. After completing this question and before continuing in the text, we suggest that you develop a model for the motion of the spring. The next section completes such a model.

- Give the three lengths that sum to the total length of the spring.
- Give the force(s).
- Besides these lengths and forces, give other variables and constants for the model.

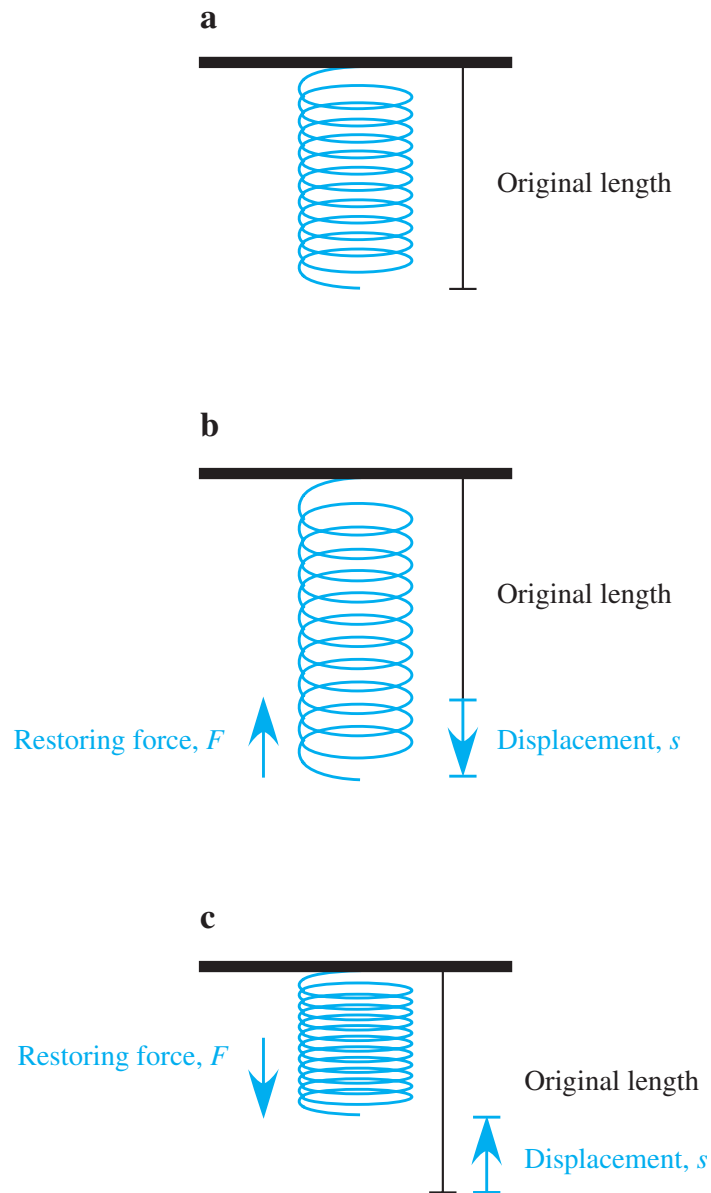


Figure 3.2.1 Original, stretched, and compressed spring

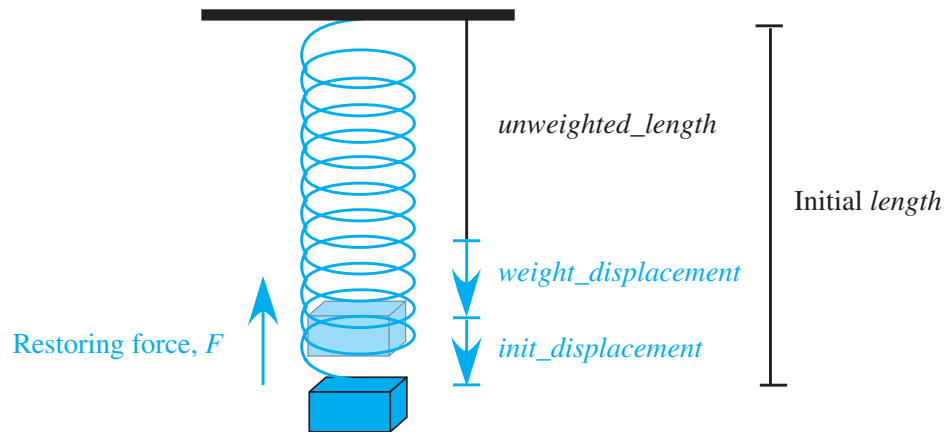


Figure 3.2.2 Vertical spring with attached weight

Vertical Springs

Before modeling bungee jumping, it is helpful to examine the action of a vertical spring, such as in Figure 3.2.2, hanging from a horizontal surface with an attached weight. Because we are considering lengths, having the positive direction be down is convenient. The initial length (*unweighted_length*) of the spring is augmented by a displacement due to the weight (*weight_displacement*) and an additional displacement due to stretching or compressing (*init_displacement*). Thus, we initialize the length of the spring (*length*) to be the following:

$$\text{unweighted_length} + \text{weight_displacement} + \text{init_displacement}$$

We enter *unweighted_length* and *init_displacement* as parameters, but a system dynamics tool can calculate *weight_displacement* because the displacement due to weight, which is a force, conforms to Hooke's law, $F = -ks$ or $s = -F/k$. Using the variables of the model, we have the following equation:

$$\text{weight_displacement} = -\text{weight}/\text{spring_constant}$$

A system dynamics diagram for the action of a vertical spring is similar to the diagram of the motion of a ball under the influence of air friction in Figure 3.1.4 of Module 3.1, “Modeling Falling and Skydiving.” We change the name of the stock *position* to *length* and replace the section concerning air friction with one involving the force due to Hooke's law. Moreover, because down is positive, *acceleration_due_to_gravity* is $+9.81 \text{ m/s}^2$ instead of -9.81 m/s^2 , as in the skydiving module. Figure 3.2.3 presents a model diagram of the action of a vertical spring experiencing no friction, or an **undamped vertical spring**, with changes to converters/variables from Figure 3.1.4 in color. With the total displacement at any instant being

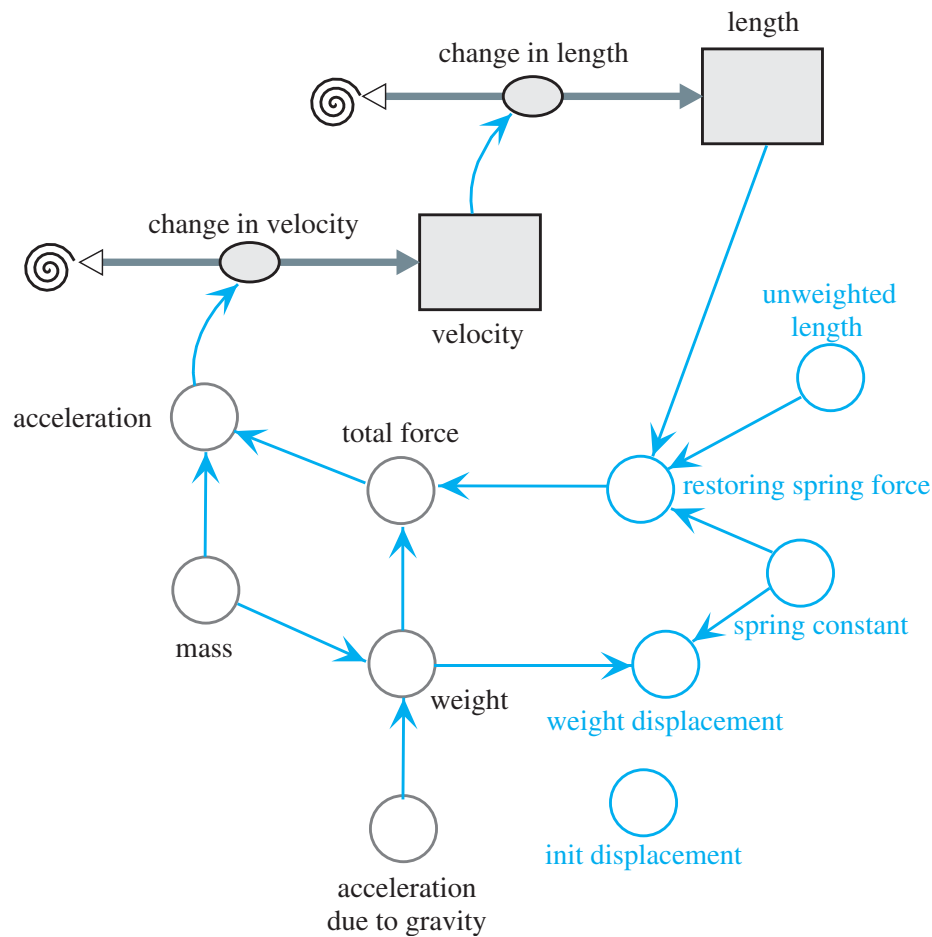


Figure 3.2.3 Model diagram of action of undamped vertical spring

$(length - unweighted_length)$, the following Hooke's law equation yields the restoring force of the spring:

$$restoring_spring_force = -spring_constant * (length - unweighted_length)$$

For a simulation, suppose we consider hanging a 0.2-kg mass on the end of a 1-m spring that has a spring constant of 10 N/m. The 0.2-kg mass exerts a force of $F = mg = (0.2 \text{ kg})(9.81 \text{ m/s}^2) = 1.962 \text{ N}$, its weight. The resulting displacement because of the weight is $1.962 \text{ N} / (10 \text{ N/m}) = 0.1962 \text{ m} = 19.62 \text{ cm}$. Thus, the length of the resting spring with an attached 0.2-kg mass is $1 \text{ m} + 0.1962 \text{ m} = 1.1962 \text{ m}$. If we then consider pulling the weight an additional $0.3 \text{ m} = 30 \text{ cm}$, the simulation with $\Delta t = 0.02 \text{ s}$ produces the graph of the length of the spring in Figure 3.2.4 with an initial length of $1 + 0.1962 + 0.3 = 1.4962 \text{ m}$. Because the spring is undamped, the simulation indicates perpetual oscillations. The length fluctuates from a maximum of 1.4962 m to a minimum of 0.8962 m . The equilibrium point, 1.1962 m , is the midway point between the two extremes and the length of the weighted motionless spring. The extremes of the oscillation are each $init_displacement = 0.3 \text{ m}$ from the

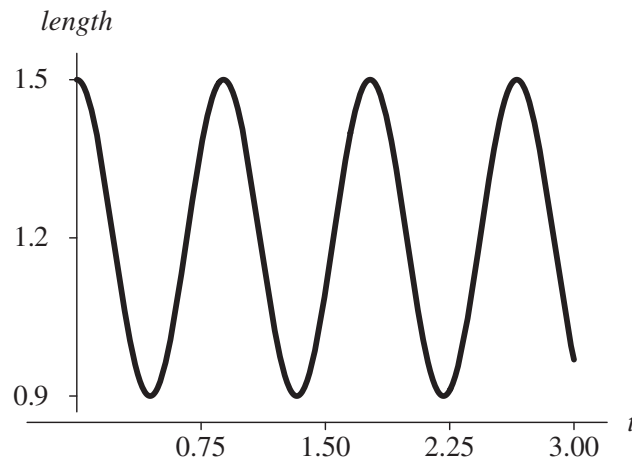


Figure 3.2.4 Graph of length (m) with respect to time (s) in undamped spring

equilibrium point. Such periodic oscillating motion is an example of **simple harmonic motion**, whose definition follows.

Definition A **simple harmonic oscillator** satisfies the following conditions:

1. The system oscillates around an equilibrium position.
2. The equilibrium position is the point at which no net force exists.
3. The restoring force is proportional to the displacement.
4. The restoring force is in the opposite direction of the displacement.
5. The motion is periodic.
6. All damping effects are neglected.

Quick Review Question 3

Suppose a weight of 8 N hangs from a spring with unstretched length of 2 m and spring constant of 100 N/m. Then, the spring is compressed 0.5 m. Using down as the positive direction and ignoring drag, give the following along with units:

- a. The displacement caused by the weight
- b. The equilibrium position
- c. The maximum length
- d. The sign of the initial restoring force
- e. The restoring force when the length of the spring is 3.15 m

Before refining the model to include friction, we should point out that a small time step, $\Delta t = 0.02$ s, resulted in Figure 3.2.4, a smooth graph that has approxi-

mately the same amplitude (distance from midpoint to high point) at each cycle. However, had we picked a time step 10 times larger, $\Delta t = 0.2$ s, then the graph would be jagged with decreasing amplitudes, both of which are artifacts of the simulation instead of reflections of reality. Too large a time step can produce misleading results that can bring about faulty conclusions.

Rule of Thumb Although not always incorrect, we should be suspicious that a time step is too large if a resulting graph is not smooth but has sharp changes in direction with what appears like line segments glued together.

Returning to our model, a damped spring, which does experience friction, also is a simple harmonic oscillator. The diagram is as in Figure 3.2.3, with the addition of another force, **drag**, which becomes part of the total force. Exercise 5 and projects refine the model to include friction. The resulting graph in Figure 3.2.5 shows a damped oscillation with the same period as the corresponding spring with no friction.

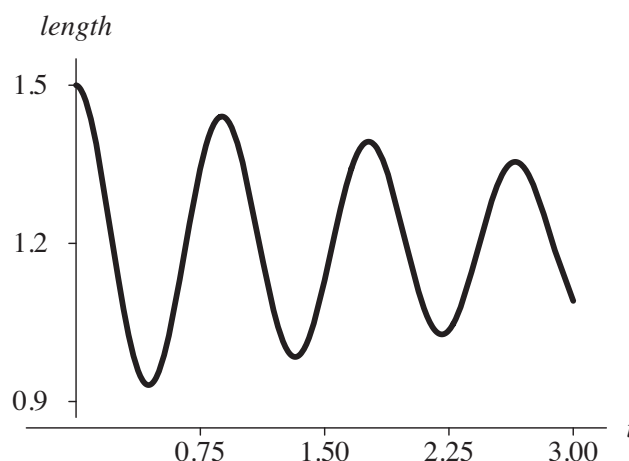


Figure 3.2.5 Graph of length (m) with respect to time (s) in damped spring

Modeling a Bungee Jump

A model of a bungee jump is very similar to that of the weighted spring under the influence of air friction. As a simplifying assumption, we ignore the weight of the bungee cord. Thus, the forces are the weight of the jumper, the restoring force of the cord, and air resistance. The only difference between this model and that of a weighted damped spring is in the equation for *restoring_spring_force*. A vertical spring is fairly rigid, so that when the weight is above the equilibrium point, the spring exerts a restoring force in the opposite direction. However, when the bungee

jumper is above the unweighted length, the cord is slack and, we can assume, exerts no downward force. This unweighted length is the length of the original, unstretched cord without the jumper. Thus, the value of *restoring_spring_force* is a conditional expression. If the length is stretched beyond *unweighted_length*, then the restoring spring force obeys Hooke's law with the displacement being (*length* - *unweighted_length*). When the jumper flies above *unweighted_length*, the bungee cord does not exert such a force. Thus, the value of *restoring_spring_force* has the following logic:

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if (length > unweighted_length) then
    restoring_spring_force  $\leftarrow$  -spring_constant * (length - unweighted_length)
else
    restoring_spring_force  $\leftarrow$  0

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Figure 3.2.6 presents a diagram for the motion of a bungee jumper with *total_force* summing *weight*, *air_friction*, and *restoring_spring_force* and with the additional converters/variables from Figure 3.2.3 in color. Figure 3.2.7 displays graphs for the bungee cord's length and the jumper's velocity for a cord with spring constant 6 N/m, unweighted length 30 m, and initial length, or distance from the top of the bridge, of 0 m. For this simulation, the jumper has mass 80 kg (equivalent to a

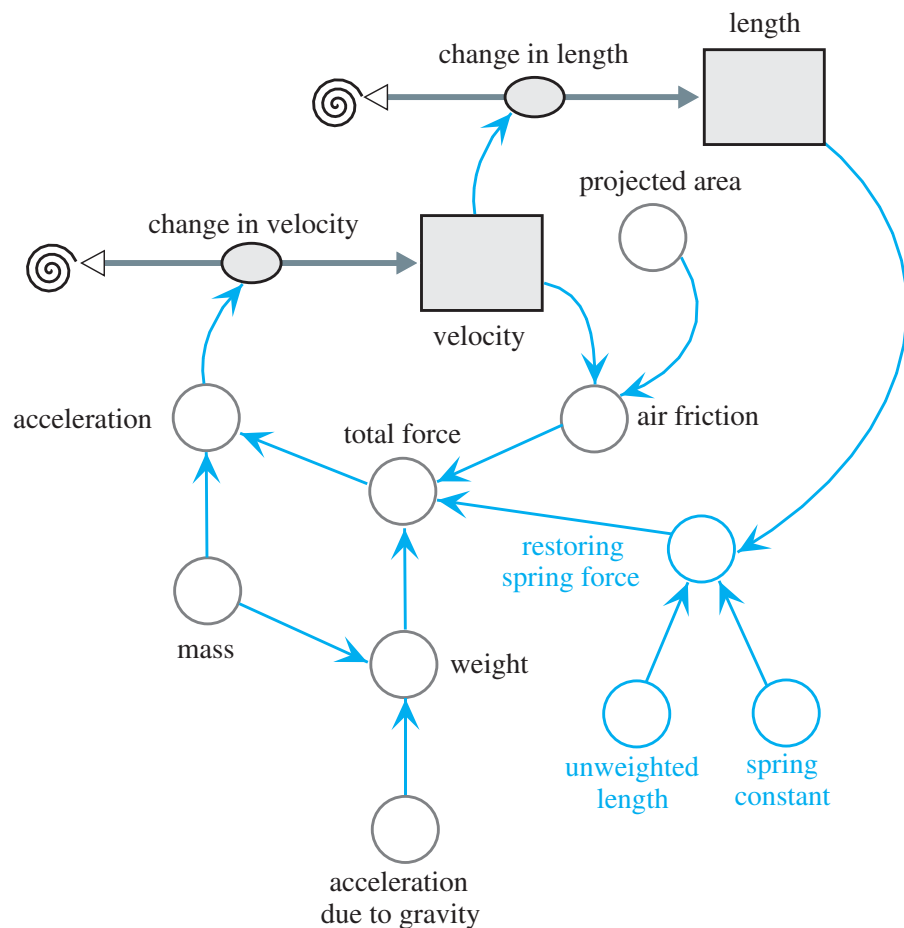


Figure 3.2.6 Diagram for motion of bungee jumper

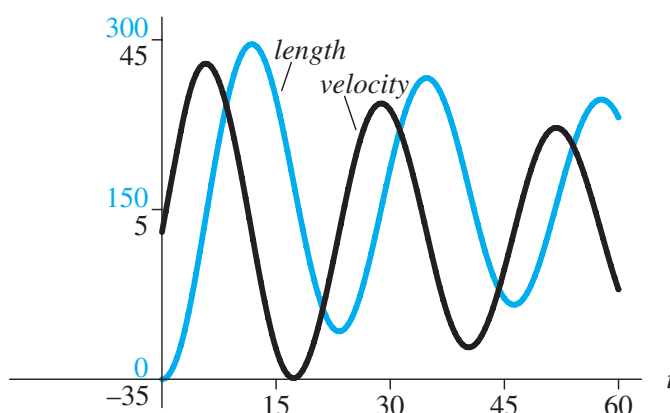


Figure 3.2.7 Graphs of length (m) and velocity (m/s) for bungee jump

weight of about 176.4 lb) and by jumping headfirst, has a small projected area of about 0.1 m^2 . The graphs show the simple harmonic oscillation and the damping motion due to drag. One of the projects explores choosing an appropriate bungee cord for a jumper.

Quick Review Question 4

With down being the positive direction, give the formulas for each of the following components of the diagram for a bungee jump (Figure 3.2.6).

- total_force*
- acceleration*
- weight*
- acceleration_due_to_gravity* with two digits after the decimal
- air_friction* assuming Newtonian friction, sea-level air density, and absolute value function ABS
- restoring_spring_force* when *length* is greater than *unweighted_length*
- restoring_spring_force* when *length* is less than *unweighted_length*

Exercises

- Give the changes in the diagram and equations for the model of the damped spring to account for air friction.
- Write a differential equation in terms of displacement s to model the motion of an undamped vertical spring with mass m . Recall that weight should equal the spring's restoring force. Use the following variables: s , displacement; t , time; m , mass; k , spring constant.
 - Using calculus, show that $s(t) = c_1 \cos\left(\sqrt{\frac{k}{m}}t\right) + c_2 \sin\left(\sqrt{\frac{k}{m}}t\right)$ is a solution to the differential equation.

- c. From Part b, determine the period of the vibrations.
- d. Using $k = 10 \text{ N/m}$ and $m = 0.2 \text{ kg}$, determine the period of the vibrations. Does your answer agree with the graph in Figure 3.2.4?
- 3. Write a differential equation in terms of displacement s to model the motion of a damped vertical spring with mass assuming Newtonian friction.
- 4. Write a differential equation in terms of displacement s to model the motion of a damped vertical spring with mass assuming Stokes' friction.
- 5. Write differential equations in terms of displacement s to model the motion of a bungee jumper. The system should model two situations, $s \geq \text{unweighted_length}$ and $s < \text{unweighted_length}$.

Projects

For additional projects, see Module 7.8, "Electrical Circuits—A Complete Story."

From the text's website, download a VerticalSpring file with the model of the motion of an undamped vertical spring for Projects 1–7. Download a Bungee file with the model of the motion of a bungee jump for Project 8.

1. Refine the model of the motion of an undamped vertical spring to account for air friction, and include a graph of *velocity*.
 - a. Determine if changing any of the following affects the period; and if it does, determine a relationship between the parameter and the period: *init_displacement*, *mass*, *spring_constant*.
 - b. How does changing *spring_constant* affect the graph of *length*?
 - c. Give the relationship between the length of the spring and its velocity. For example, when the velocity is zero, what is the position of the spring? At what stage(s) does the spring have a maximum velocity?
2. Refine the model of the motion of a spring to account for air friction. Run an experiment with a real spring and mass to determine the lengths at various times and the period. Estimate the spring constant using a system dynamics model.
3. Refine the model of the motion of a spring to account for drag using Stokes' friction. Suppose b is the constant of proportionality in Stokes' friction; define $u = b/(2m)$ for mass m . Also, define $w^2 = k/m$, where k is the spring constant. Show that when $u^2 > w^2$, with the damping coefficient large in comparison to the spring constant, the system is **overdamped** and displays no oscillation. However, when $u^2 < w^2$, the system is **underdamped** and does show oscillatory behavior.
4. The restoring force for a nonlinear hard spring is $ks(1 + a^2s^2)$, where k is the spring constant, s is the displacement, and a is a small constant. The restoring force for a nonlinear soft spring is $ks(1 - a^2s^2)$. Develop models for such springs. Discuss and compare their motions.
5. Model the motion of an aging spring by replacing the spring constant k with a decreasing function ke^{-at} , where a is a positive constant and t is time.
6. Adjust the equations of the spring model to compute the spring constant given a weight and a corresponding displacement.

7. Model a damped oscillator with parameters very close to those of the bungee jump. Discuss the results in comparison to those of the bungee jump.
8. A bungee jumper wants to have a “great ride,” getting close to the ground without hitting it. Suppose the distance of the jumping bridge above a gorge is 80 m and the length of the cord is 30 m. Determine the most appropriate whole number spring constants for jumpers of masses 60 kg, 70 kg, and 80 kg. Employ comparison graphs. Discuss your results.
9. The **buoyancy** of a floating object is the restoring force to return the object to its normal floating layer after a vertical displacement. This force is equal to $-grA^s$, where g is the acceleration due to gravity, r is the fluid density, A is the cross-sectional area of the object, and s is the displacement from the normal floating layer. Design a model for the motion of a displaced object, and discuss the results of the simulation. Let the density of water be 1 g/cm^3 .

Answers to Quick Review Questions

1. a. N/m or kg/s^2 because $k = -F/s$, s is in m, and F is in N or kg m/s^2
b. $-(5 \text{ kg/s}^2)(0.1 \text{ m}) = -0.5 \text{ N}$
2. a. resting length of spring with no weight, displacement from that length due to the weight, initial displacement due to pulling down or pushing up the weight
b. weight and restoring force of spring
c. Velocity, acceleration, mass, acceleration due to gravity, spring constant; a diagram of the model has the basic form of Figure 3.1.4 for an object falling with friction. Before continuing, we suggest that you revise this figure and include equations to model the motion of a spring, using *length* instead of *position* and ignoring friction.
3. a. $8 \text{ N}/(100 \text{ N/m}) = 0.8 \text{ m}$
b. $2 \text{ m} + 0.8 \text{ m} = 2.8 \text{ m}$
c. $2.8 \text{ m} + 0.5 \text{ m} = 3.3 \text{ m}$
d. positive
e. $-(100 \text{ N/m})(3.15 \text{ m} - 2.00 \text{ m}) = -15 \text{ N}$
4. a. *weight + air_friction + restoring_spring_force*
b. *total_force/mass*
c. *mass * acceleration_due_to_gravity*
d. 9.81
e. $-0.65 * \text{projected_area} * \text{velocity} * \text{ABS}(\text{velocity})$
f. $-\text{spring_constant} * (\text{length} - \text{unweighted_length})$
g. 0

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